ROBERT BAIN

COMPUTER VISION ENGINEER | MACHINE LEARNING | SEMICONDUCTOR R&D

Irvine, CA | Relocating to Hillsboro area, Oregon | rbain18@gmail.com | rkbain.com | github.com/bainro

ENGINEERING PROFILE

Computer vision and machine learning engineer with an M.S. in Artificial Intelligence, production software experience, and hands-on semiconductor cleanroom fabrication. Developed real-time perception for autonomous vehicles, GPU-based CNN research, multi-camera imaging tools, and a complete neural recording platform from photomasks through electronics and analysis. Originally from Newberg, Oregon and relocating to the Portland metro area.

TECHNICAL STRENGTHS

Computer vision: YOLO object detection, camera calibration, stereo 3D tracking, 3D pose estimation, ResNet, CNN regression, visual localization, image processing, geometric CV

ML and software: Python, C++, CUDA, PyTorch, TensorFlow, OpenCV, MATLAB, SQL, JavaScript, Linux, Git, ROS, GPU programming, data pipelines

Semiconductor and hardware: Photolithography, cleanroom processing, photomask design, thin-film polymers and metals, PCB design, Intan electronics, KiCad, Fusion 360, AutoCAD

ENGINEERING AND RESEARCH EXPERIENCE

Graduate Researcher, Neural Engineering and Computer Vision | UC Irvine (2023 to 2026)

- Invented and fabricated high-density 64-channel flexible neural probes using lithography and thin-film polyimide, titanium, and gold. Reduced the manufacturing flow from four photomasks to two and developed wafer-scale fabrication recipes.
- Designed custom 128-channel acquisition headstages, a 256-channel freely moving animal recording platform, connectors, mechanical packaging, and PCB layouts. Integrated microfabrication, electronics, data acquisition, and software.
- Built Python and GPU-based analysis workflows for large electrophysiology recordings. Collected chronic datasets with approximately 120 identifiable neurons tracked for three months.
- Developed and documented manufacturing workflows, coordinated photomask and PCB suppliers, and mentored five undergraduate researchers in probe assembly, testing, and troubleshooting.

Computer Vision and Machine Learning Research | UC Irvine (2024 to 2026)

- Co-developed pairwise convolutional learning for image regression and classification. Benchmarked ResNet-10 models on UTKFace and STL-10, including low-label training, augmentation, and reference-based inference.
- First author of a 2026 submission to Machine Learning with Applications. Reported 3.64-year mean absolute error for UTKFace age estimation using a lightweight CNN trained on one RTX 2080.
- Co-authored 3D structural MRI image-analysis research using CNN ranking, volume-to-slice processing, and region-level ablation for visuospatial behavioral prediction.
- Built multi-camera video capture and calibration code and integrated DeepLabCut 3D reconstruction into a rodent pose-tracking workflow.

Computer Vision Engineer, Autonomous Racing | Global Formula Racing, Oregon State (2019 to 2021)

- Built real-time cone detection and geometric perception for an autonomous racing car. Trained deep vision models and applied homographies and Perspective-n-Point to recover scene geometry.
- Developed neural network inference workflows for constrained onboard compute and collaborated on ROS-based vehicle systems.

ADDITIONAL SOFTWARE AND AI EXPERIENCE

Robotics Researcher | Cognitive Anteater Robotics Lab, UC Irvine (2022 to 2023)

- Trained a PyTorch ResNet-18 model to infer robot position from camera images. Built ROS navigation tools for a Toyota Human Support Robot and Clearpath Jackal.
- Worked on CARLsim6, a C++ and CUDA spiking neural network simulator, and developed Git-based continuous integration workflows.

AI Researcher, DARPA Machine Common Sense | Oregon State (2020 to 2021)

- Led development of Theory-of-Mind agent components for DARPA evaluations of intuitive reasoning and task planning. OSU methods placed first in two of three evaluated categories.

Lead Software Developer | Northwest Media, Eugene, Oregon (2018 to 2020)

- Led development of web software serving approximately 150,000 users and public-sector clients. Supervised two junior developers and modernized 68 interactive training courses from Flash to HTML5.

EDUCATION

University of California, Irvine: M.S., Neurobiology and Behavior, September 2026. GPA: 4.0

Oregon State University: M.S., Artificial Intelligence, 2022. GPA: 3.6

Oregon State University: B.S., Biology, Chemistry minor, 2016

SELECTED PAPERS AND CODE

- **Pairwise Learning Increases Data Efficiency in Deep Neural Networks** Bain, Rashidi, Chrastil. Submitted to Machine Learning with Applications, October 2026. Link
- **Inferring Continuous Cognitive Traits from Limited Neuroimaging Data Using Pairwise Convolutional Neural Networks** Rashidi, Bain et al. 2026 manuscript, arXiv submission pending.
- **ePose: Let's Make EfficientPose More Generally Applicable** Lally, Bain, Alotaibi. arXiv 2021. Link

Relevant public implementations: Multi-camera 3D vision: github.com/bainro/mcn_dlc3d | ResNet robot localization: github.com/bainro/map_localization